ELEC 221 Lecture 18
CT/DT conversion and sampling DT signals

Thursday 12 November 2026

Announcements

- Quiz 8 on Tuesday (L17 and L18)
- Tutorial assignment 4 in Monday's tutorial

Last time

We modeled **sampling** of CT signals as multiplication of a (band-limited) signal with a periodic impulse train:

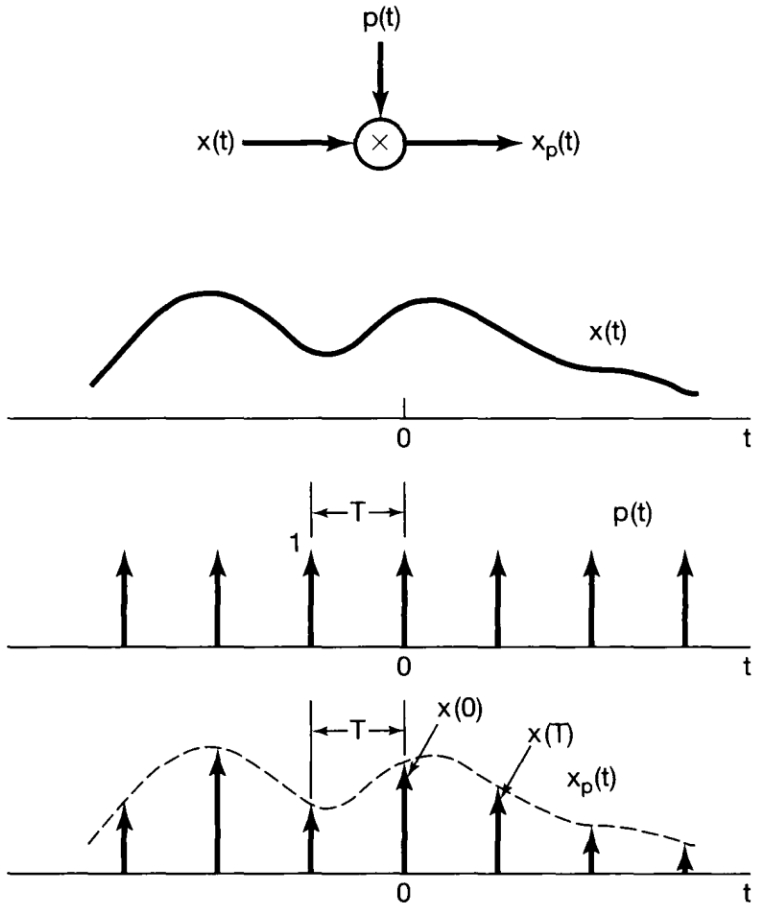


Image credit: Oppenheim 7.1

Last time

We went to the frequency domain to get a better understanding:

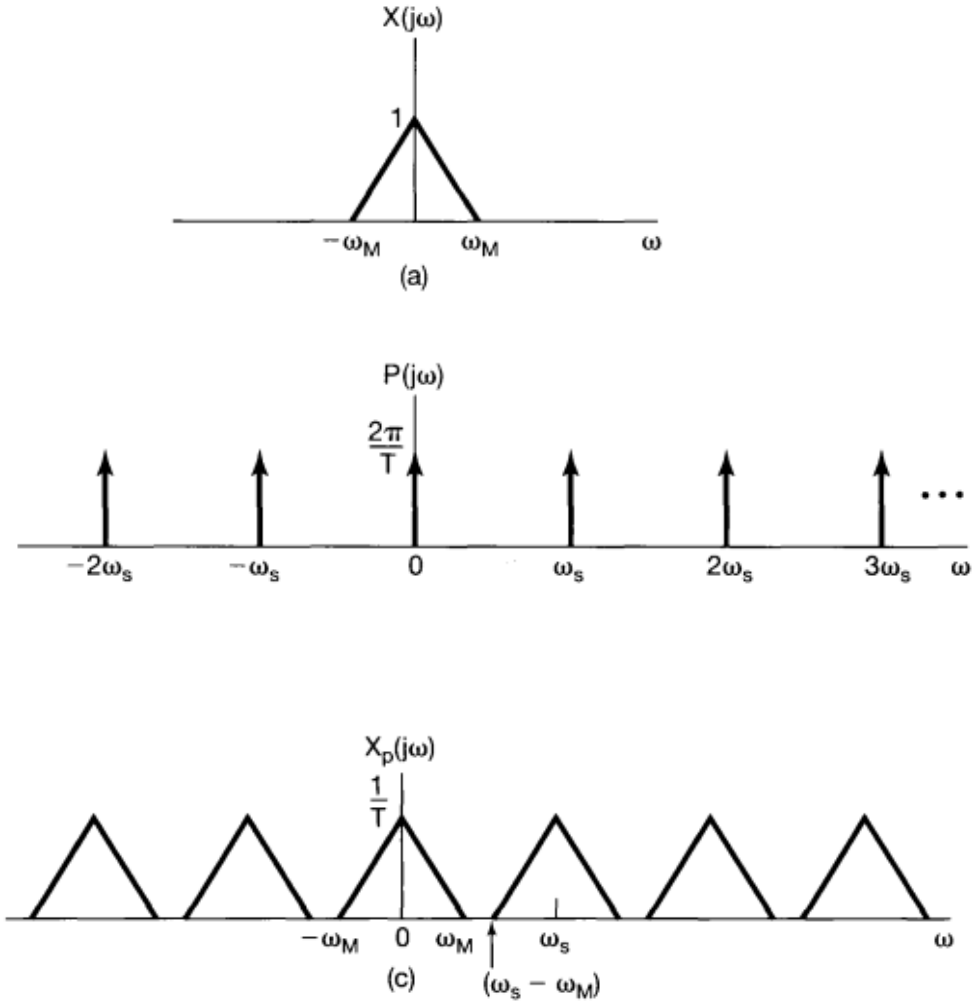


Image credit: Oppenheim 7.1

Last time

We recovered the original signal by applying a low pass filter

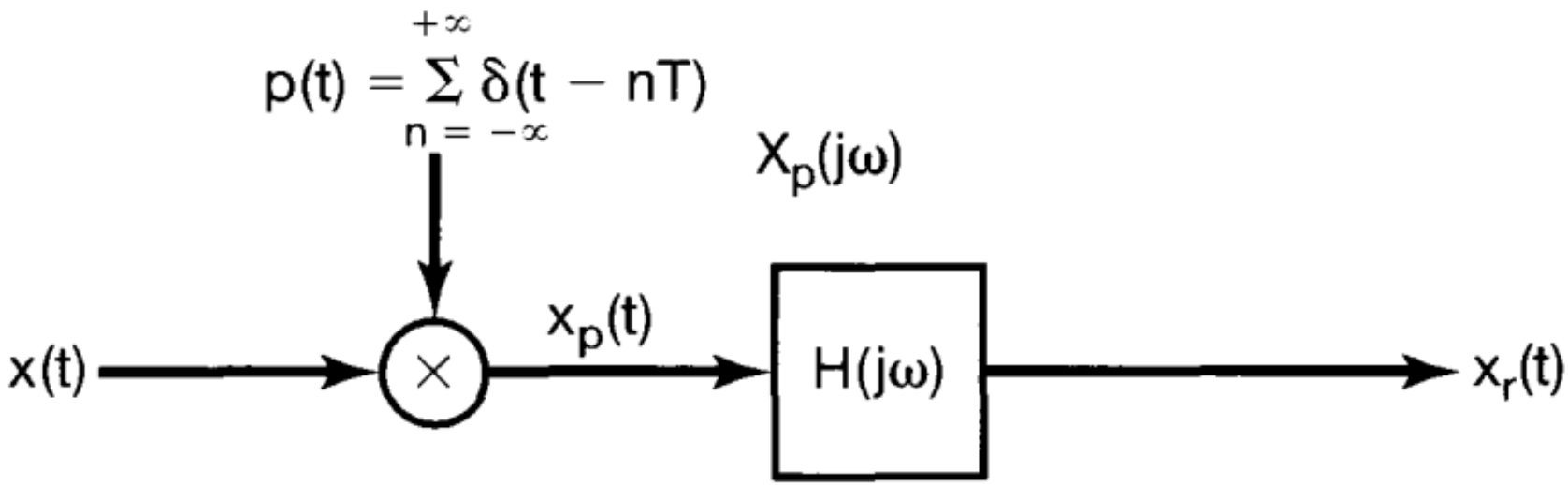
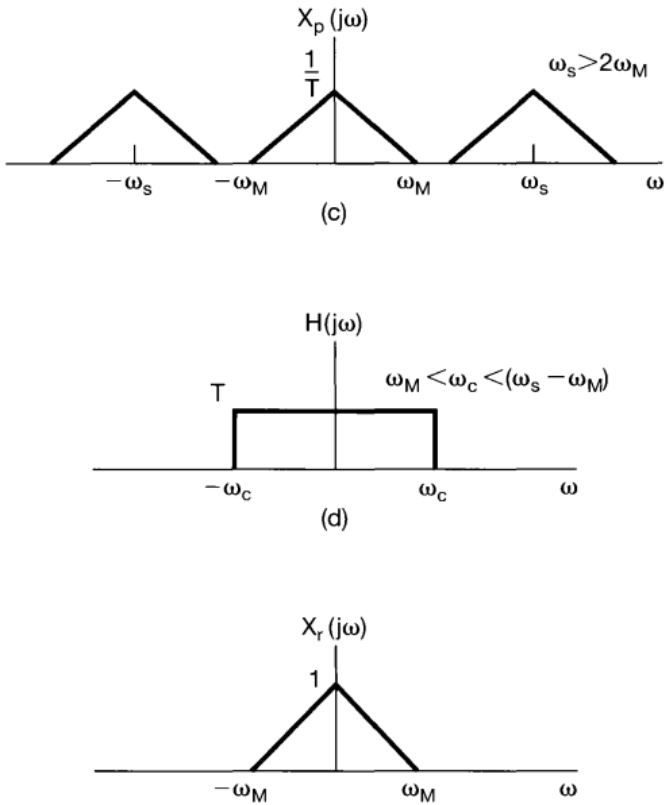


Image credit: Oppenheim 7.1

Last time



This all works if the sampling rate is higher than the **Nyquist rate**, i.e., $\omega_s > 2\omega_m$ (formally: the sampling theorem).

Image credit: Oppenheim 7.1

Today

Learning outcomes:

- describe the phenomenon of aliasing
- convert samples of CT signals to DT signals and analyze the frequency domain effects
- perform impulse train sampling on a DT signal
- decimate and interpolate a DT signal, and describe their effects on the spectrum

Aliasing

What if the sampling rate isn't high enough?

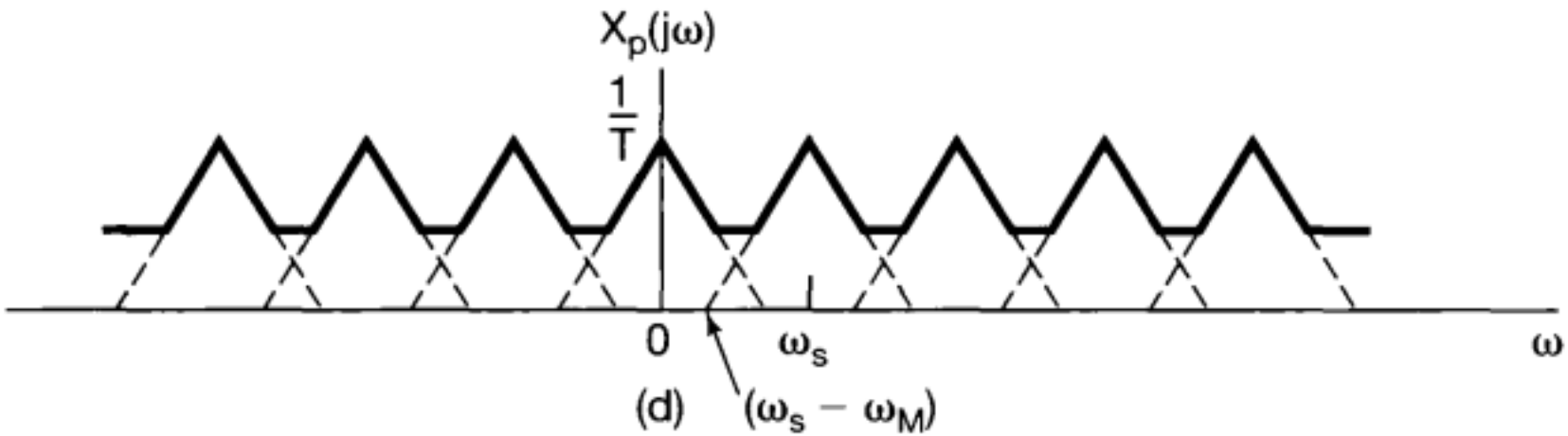


Image credit: Oppenheim 7.1

Aliasing

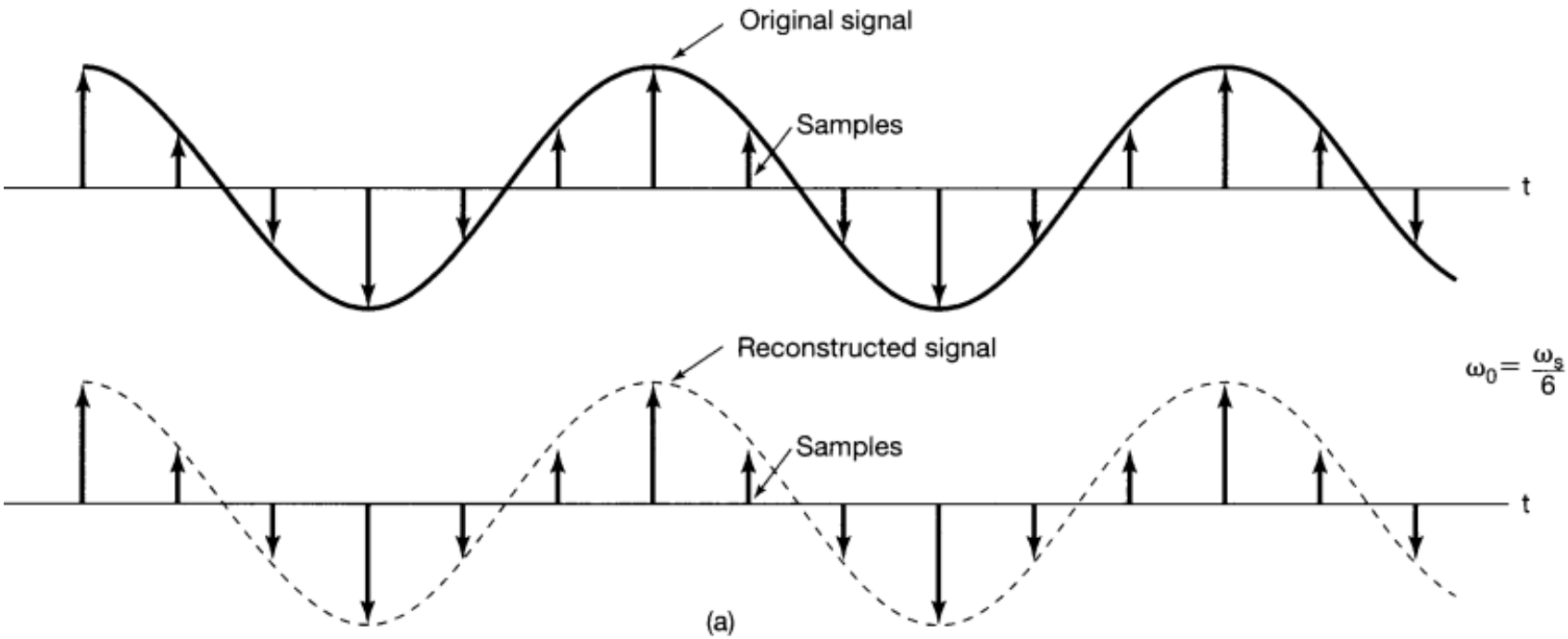


Image credit: Oppenheim 7.3

Aliasing

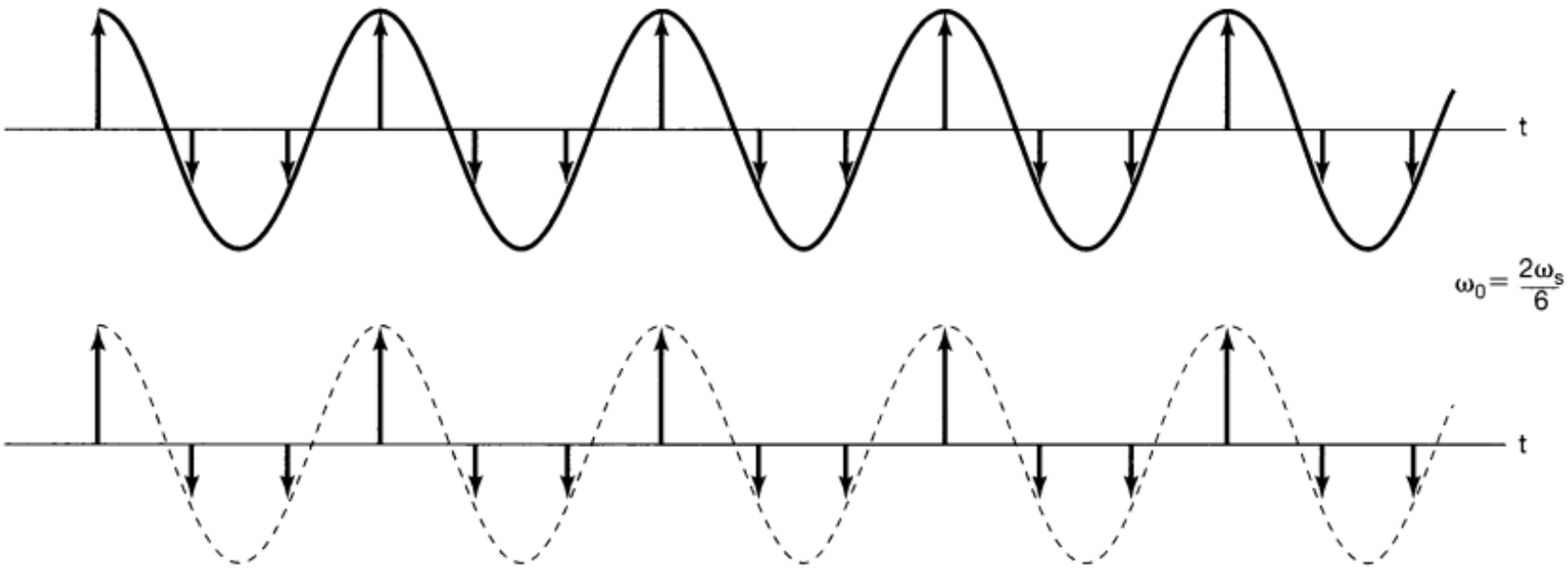


Image credit: Oppenheim 7.3

Aliasing

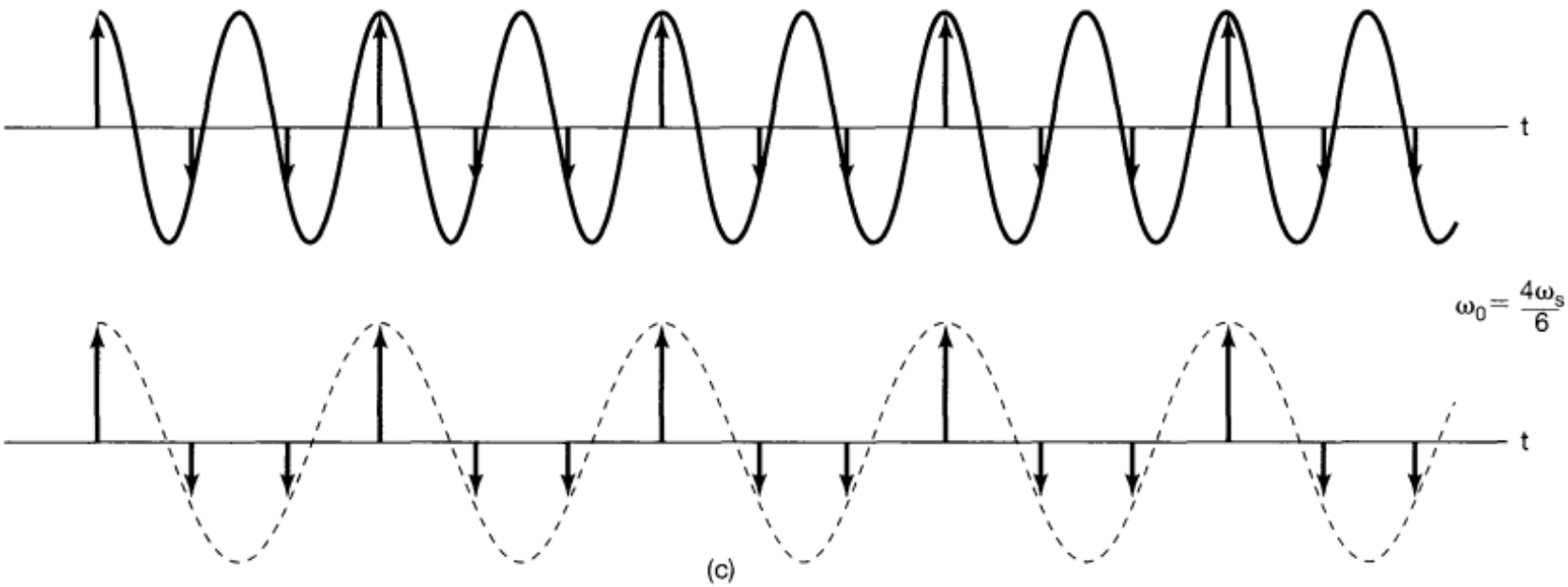


Image credit: Oppenheim 7.3

Aliasing

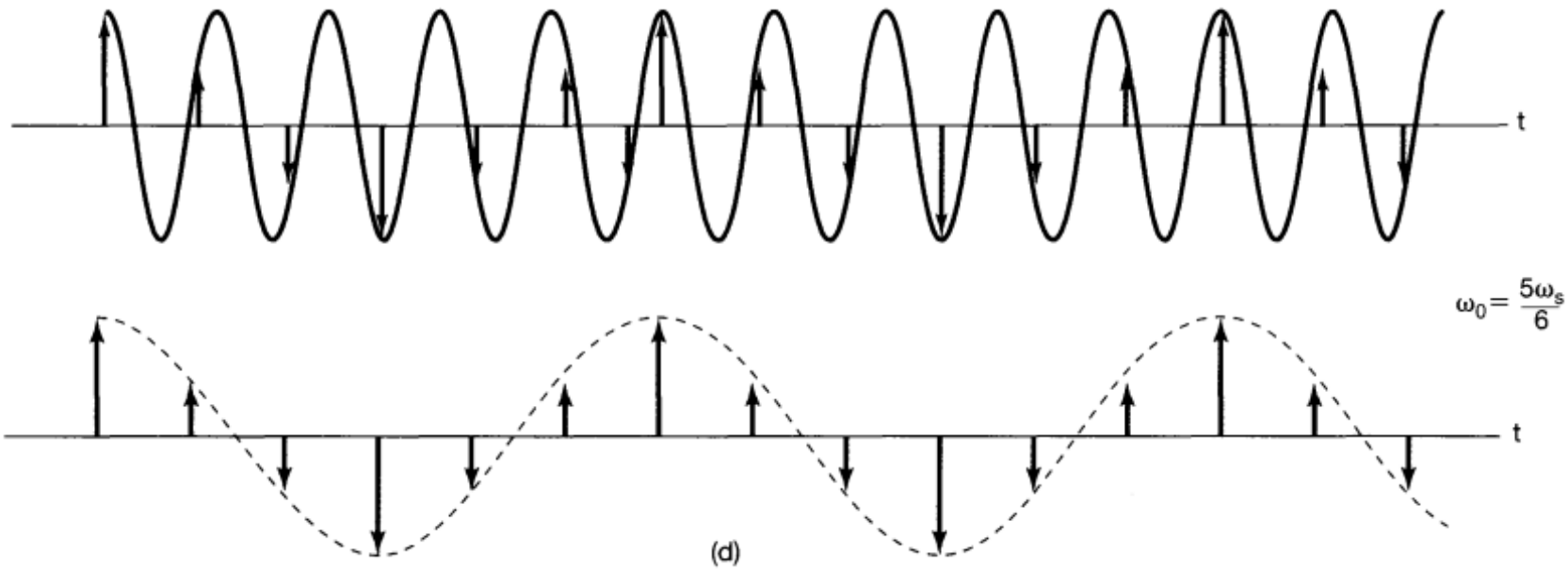


Image credit: Oppenheim 7.3

Simulation

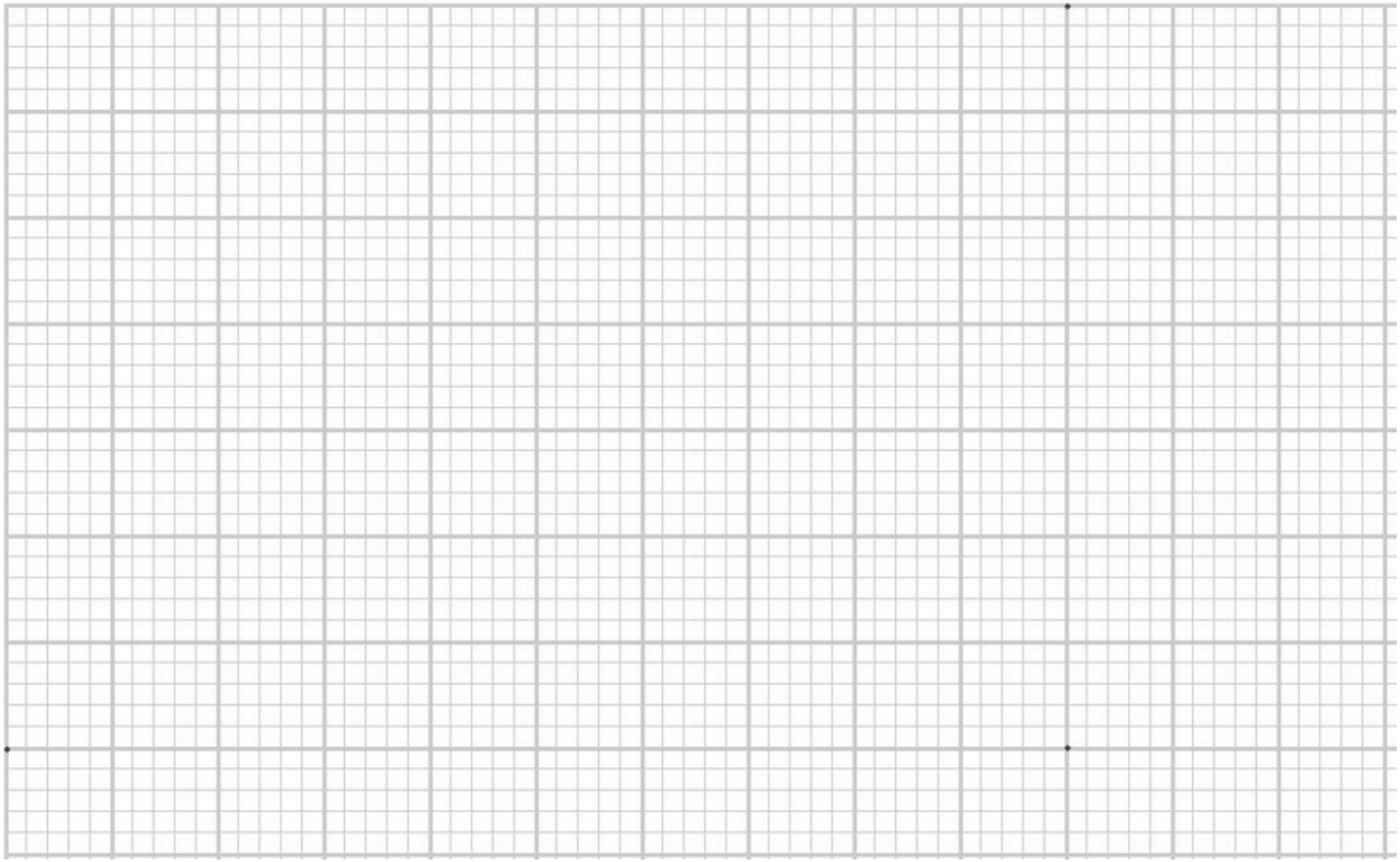
`https://visualize-it.github.io/stroboscopic_effect/simulation.html`

Two aspects to consider here:

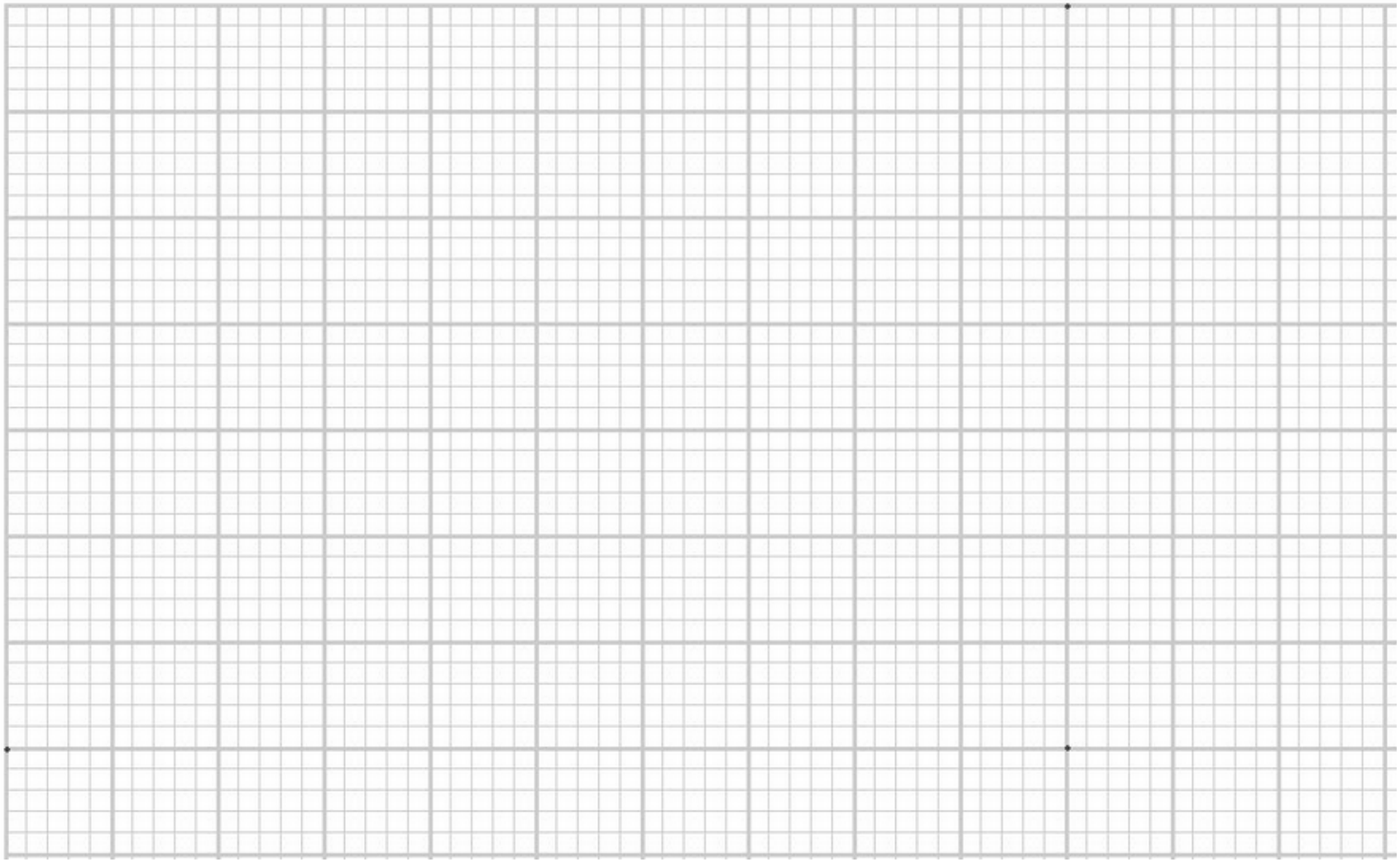
- Why does the interpreted frequency *decrease* as the true frequency increases?
- Why does it it look like it goes *backwards*?

We can understand both by looking at the spectra.

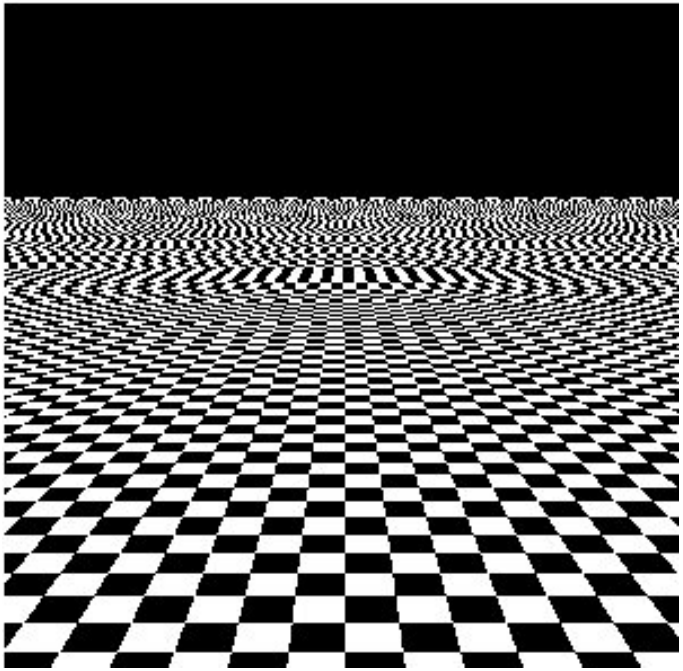
Frequency misattribution



Backwards-ness



Real-world examples



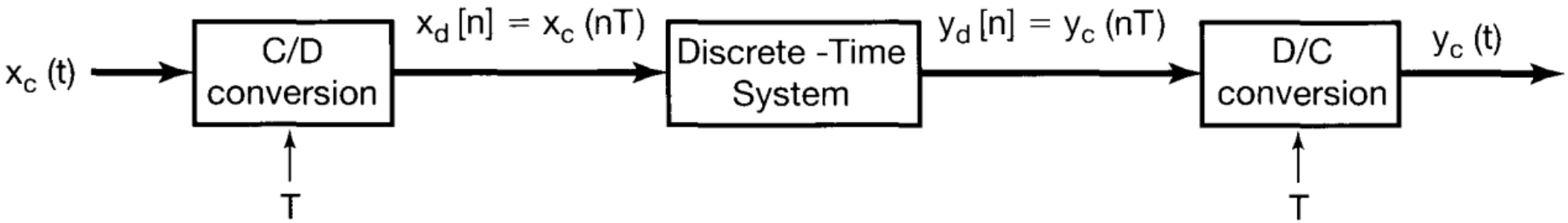
Fun on your own: read up about Moiré patterns, and various **anti-aliasing** techniques that are used in music/images/games!

Image credit: `https:`

`//textureingraphics.wordpress.com/what-is-texture-mapping/anti-aliasing-problem-and-mipmapping/`

Converting between DT $\leftrightarrow$ CT

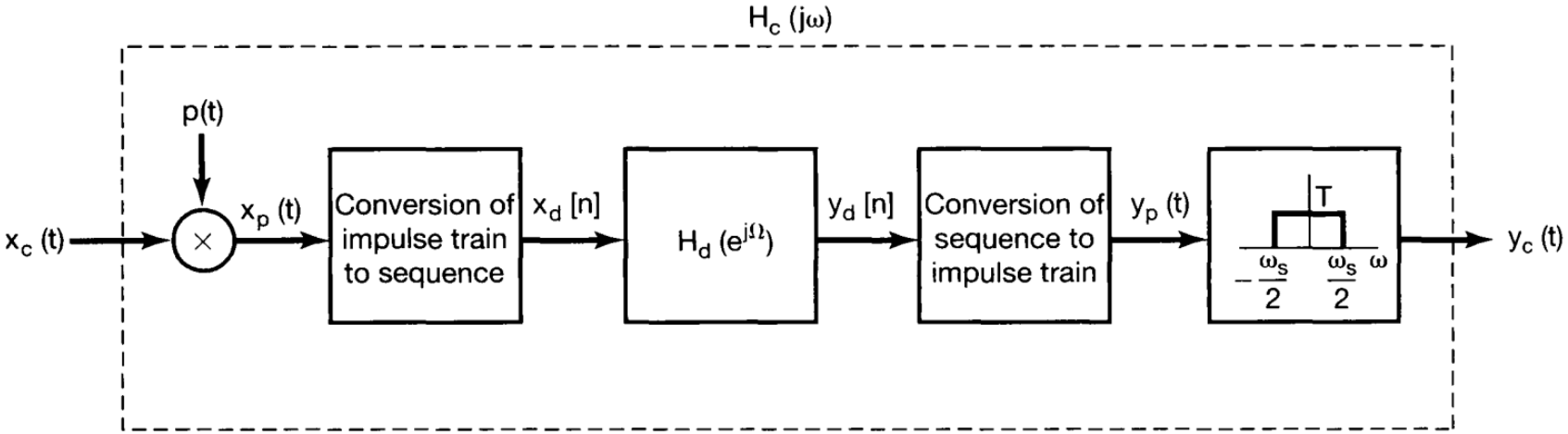
Often convenient to process CT signals by first converting to DT, processing, then converting back.



What is the theory that makes this possible?

Image credit: Oppenheim 7.20

Converting between DT $\leftrightarrow$ CT



Let's explore what happens at the level of the spectra

Note: we have *two frequencies*, one in CT, one in DT. Write:

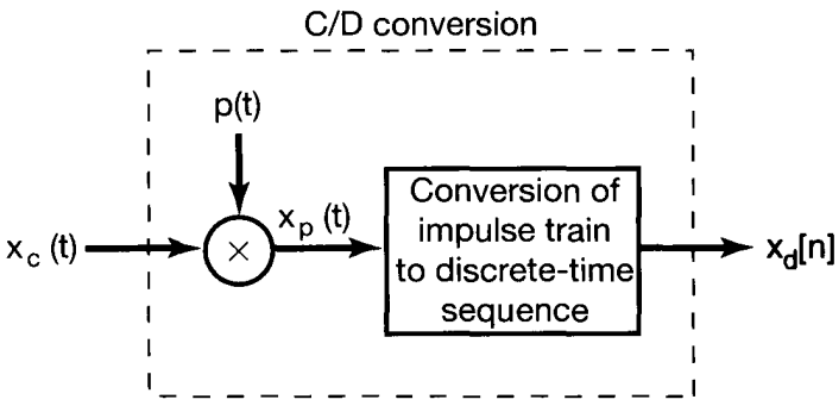
$$X_c(j\omega), \quad Y_c(j\omega)$$

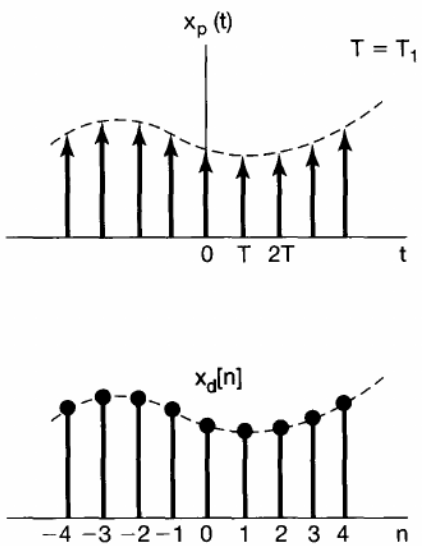
$$X_d(e^{j\Omega}), \quad Y_d(e^{j\Omega})$$

Image credit: Oppenheim 7.24

Converting between DT $\leftrightarrow$ CT

First: how are $X_p(j\omega)$ and $X_d(e^{j\Omega})$ related?

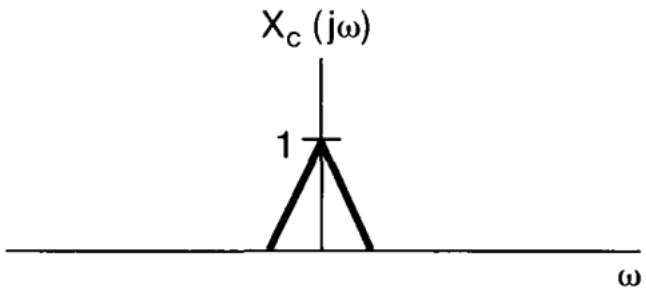


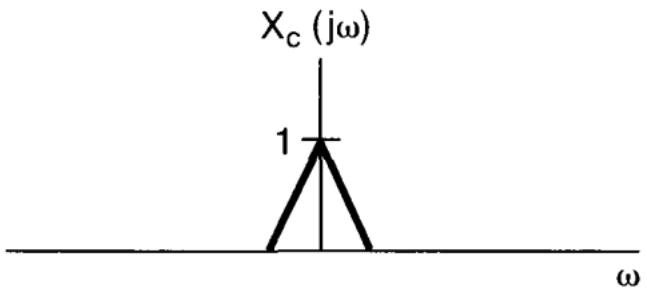


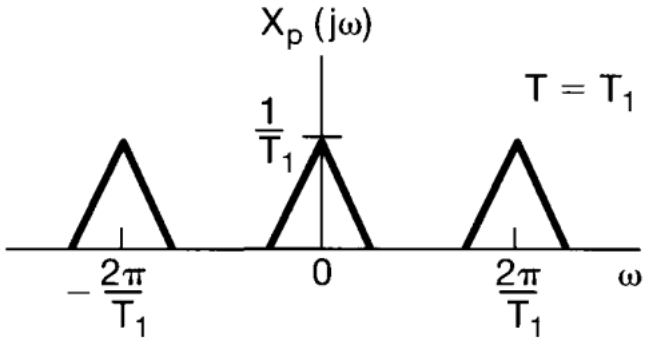
Converting between DT $\leftrightarrow$ CT

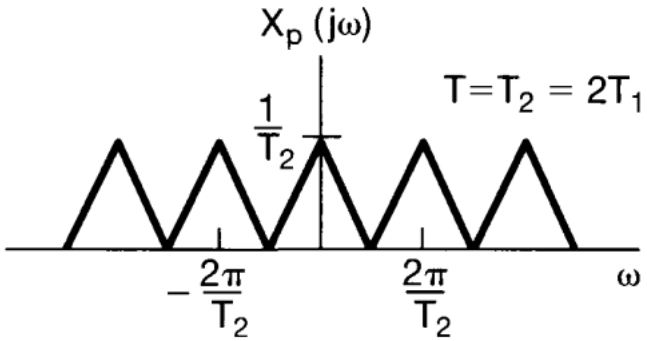
Relate $X_d(e^{j\Omega})$ back to the original spectrum $X_c(j\omega)$

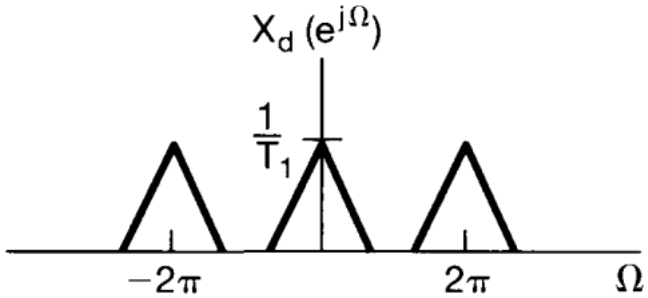
Converting between DT $\leftrightarrow$ CT

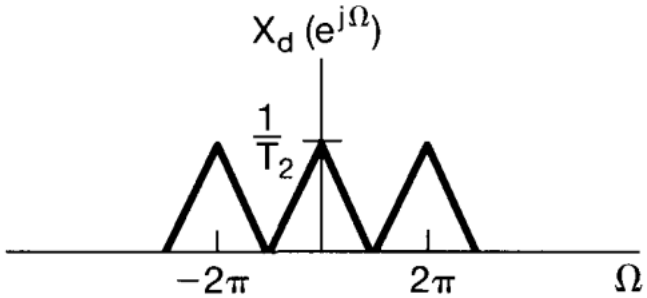




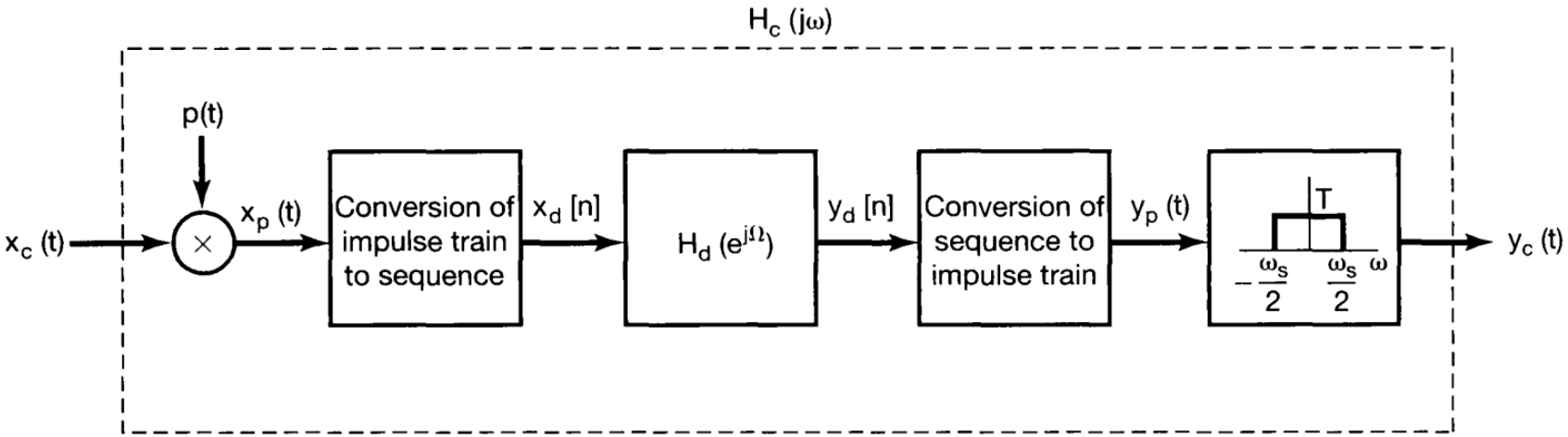








Converting between DT $\leftrightarrow$ CT



The converted signal $x_d[n]$ enters a DT system:

Image credit: Oppenheim 7.4

Converting between DT $\leftrightarrow$ CT

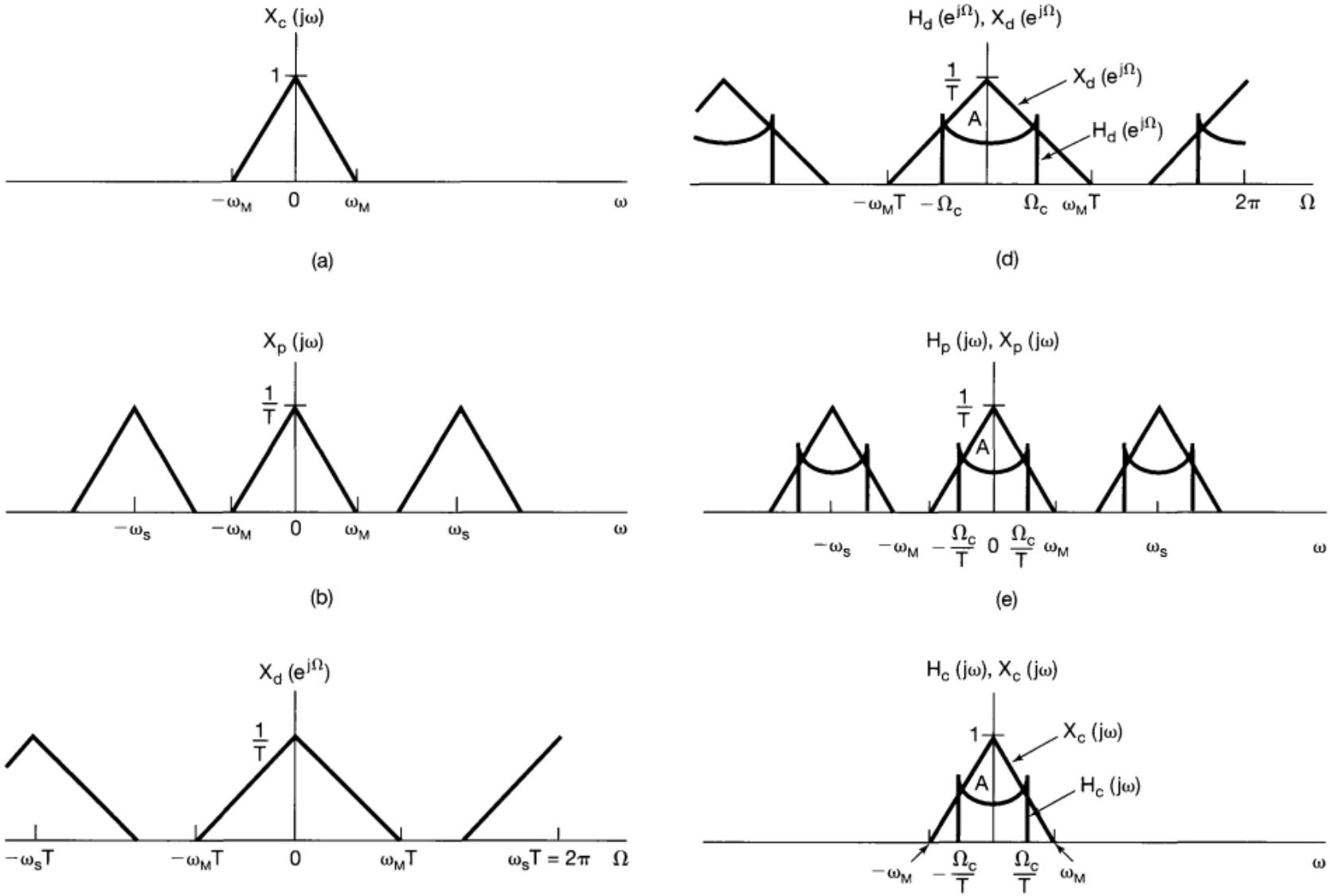
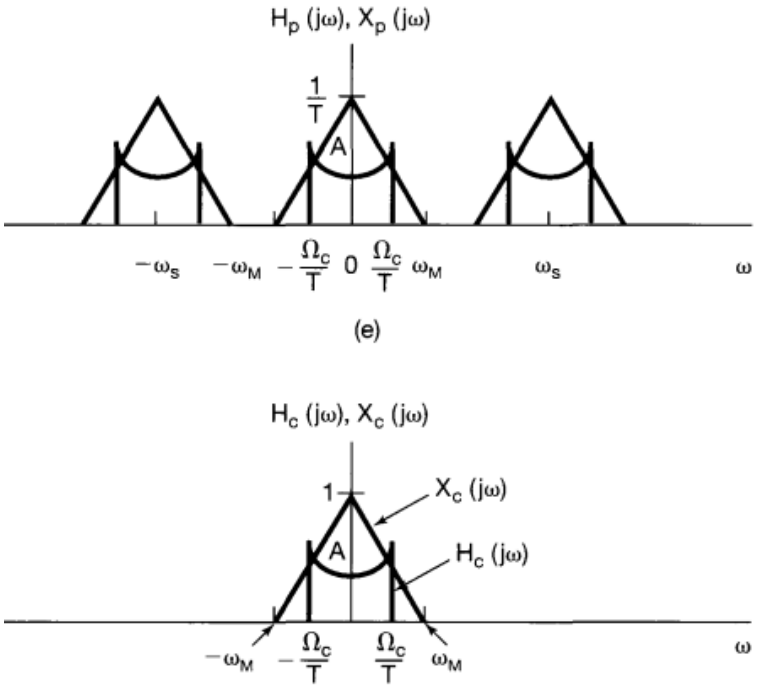
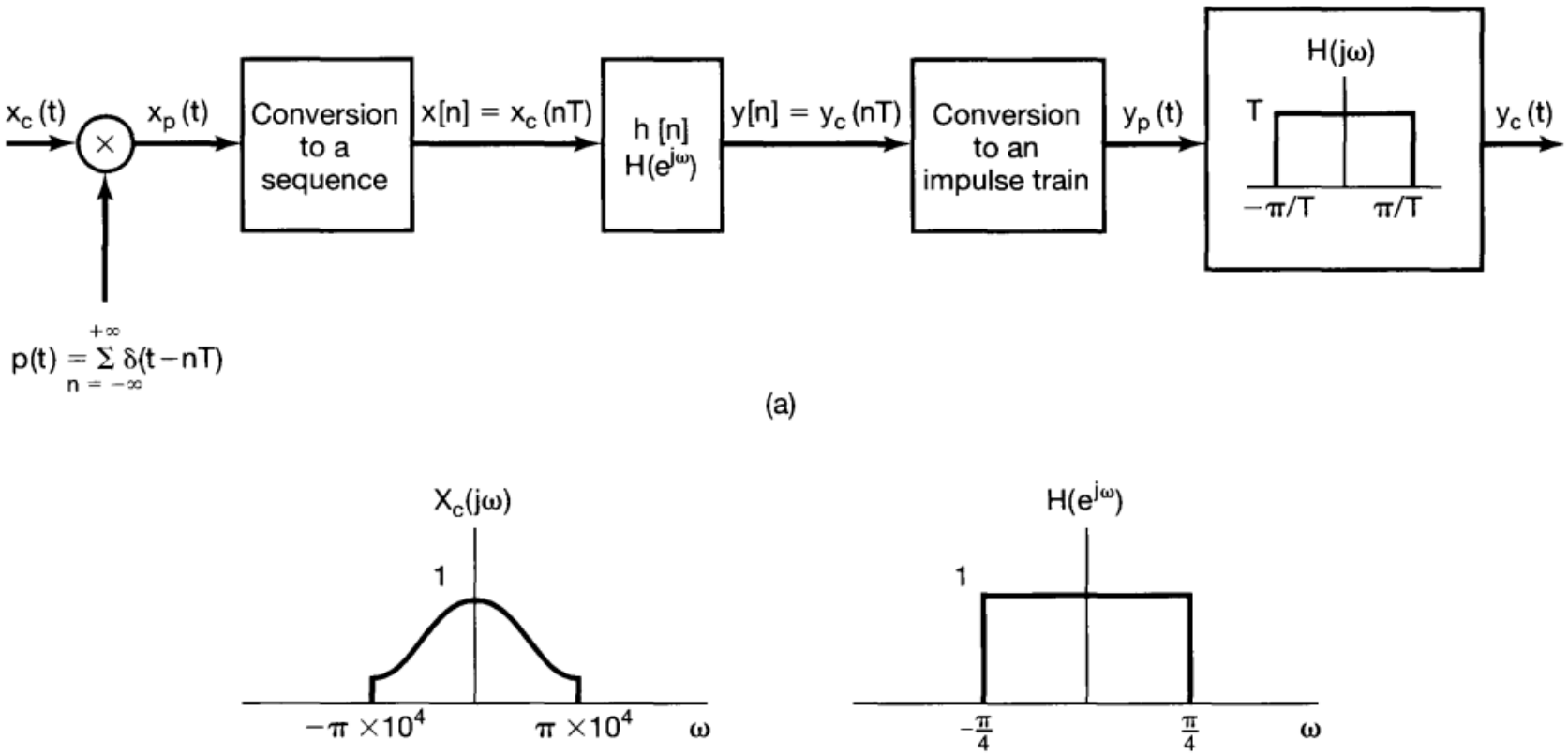


Image credit: Oppenheim 7.4

Sampling of DT signals



Example



Sketch: $X_p(j\omega)$, $X(e^{j\omega})$, $Y(e^{j\omega})$, $Y_p(j\omega)$, $Y_c(j\omega)$ if $1/T = 20kHz$.

Image credit: Oppenheim Problem 7.29

Example

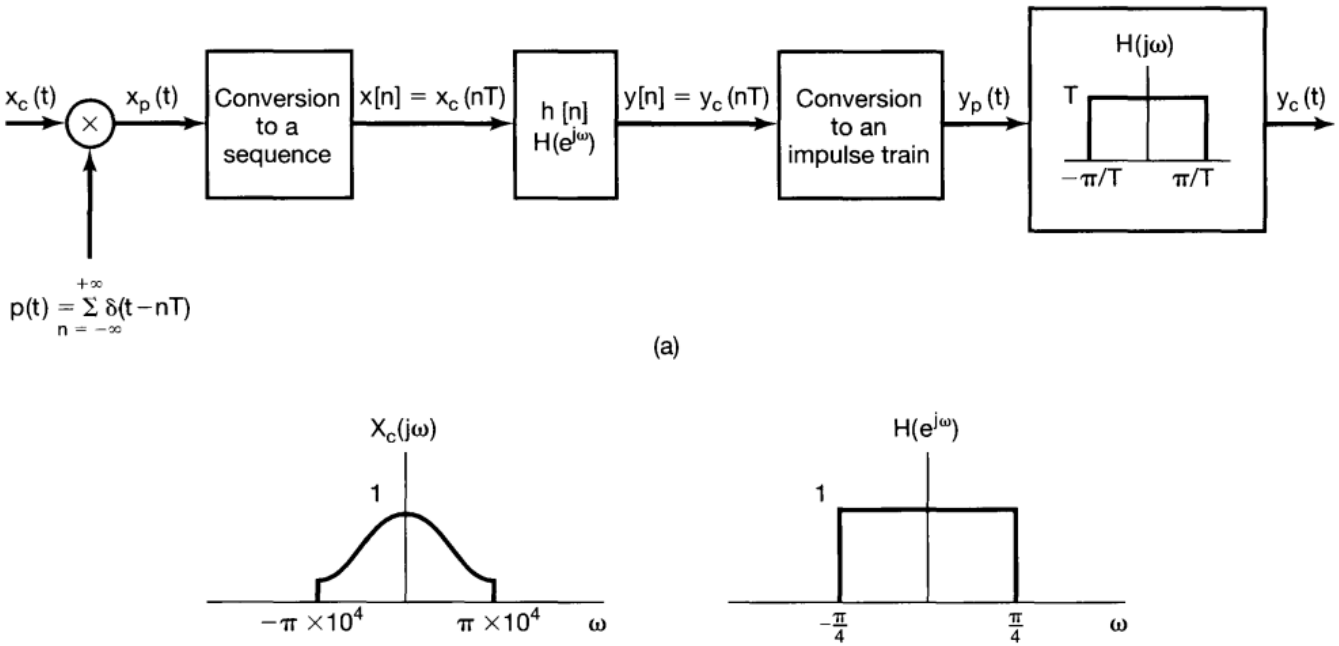


Image credit: Oppenheim Problem 7.29

Example

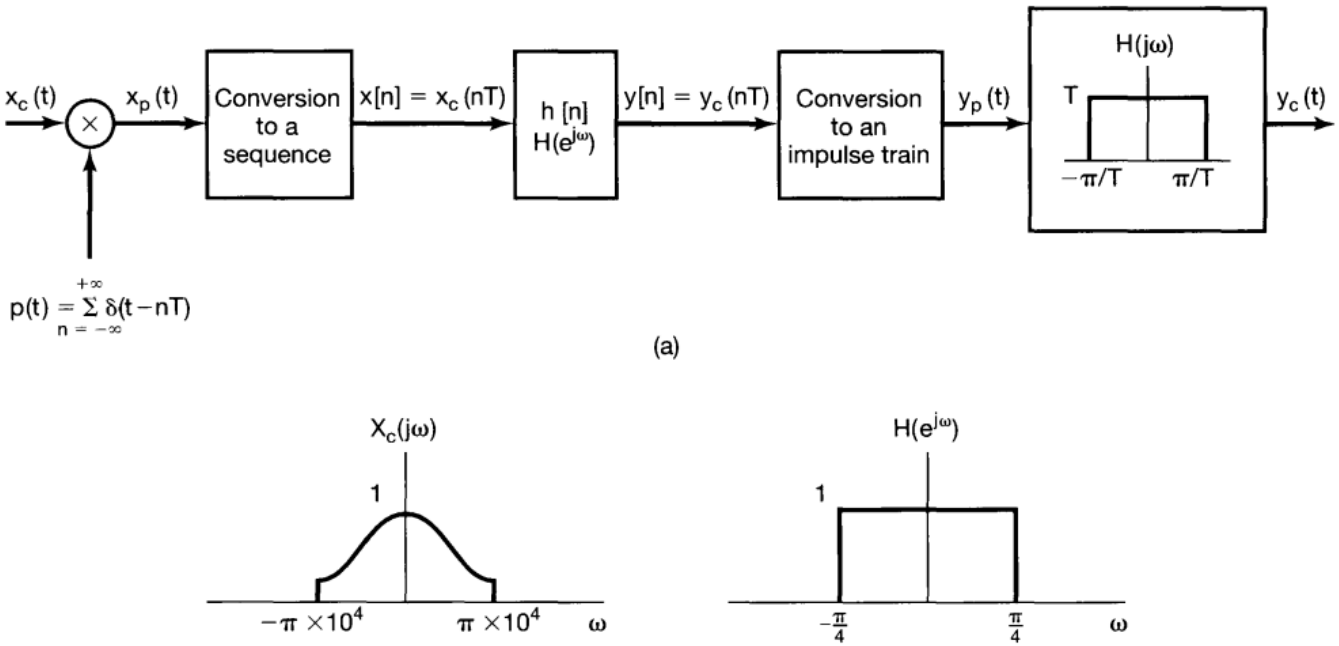


Image credit: Oppenheim Problem 7.29

Example

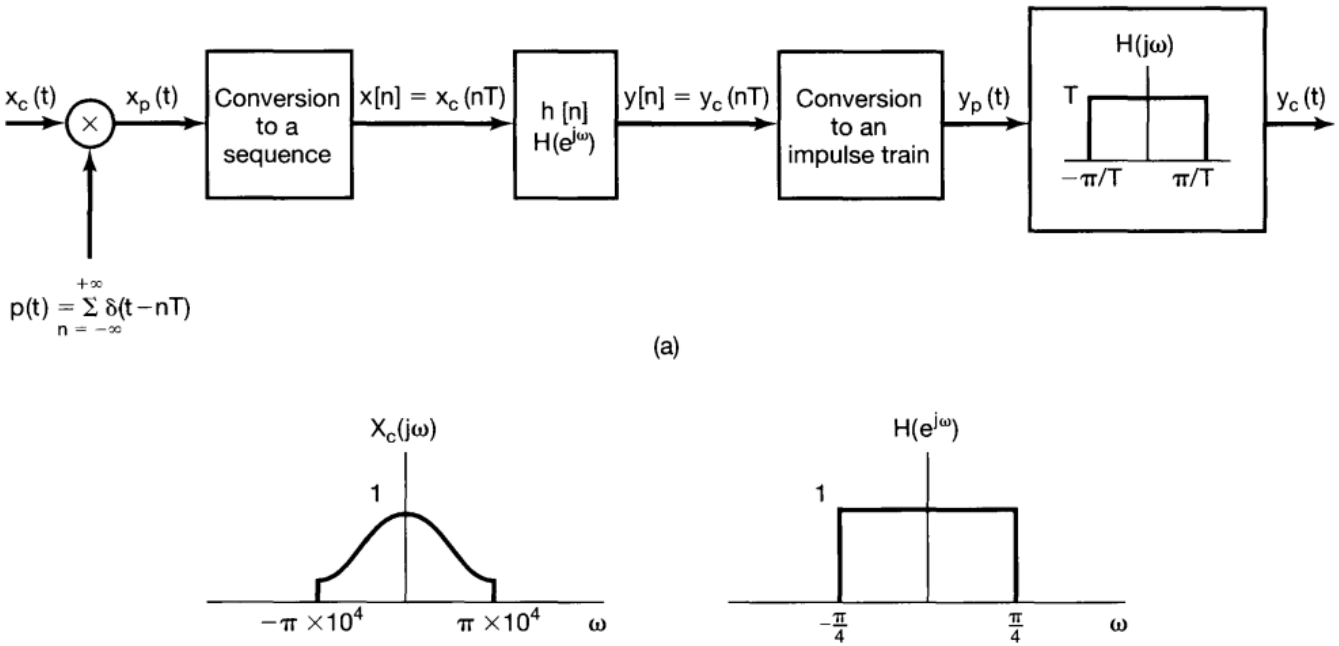


Image credit: Oppenheim Problem 7.29

Sampling of discrete-time signals

Sample with DT impulse train of period N:

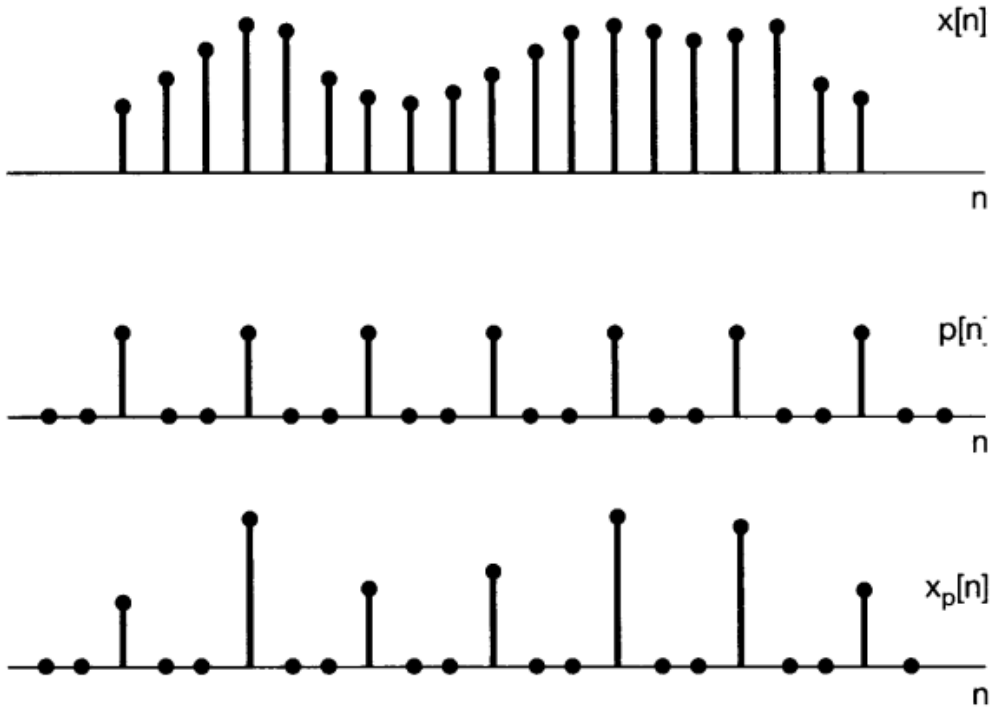


Image credit: Oppenheim 7.5

Sampling of discrete-time signals

Take similar approach as we did in CT:

Sampling of discrete-time signals

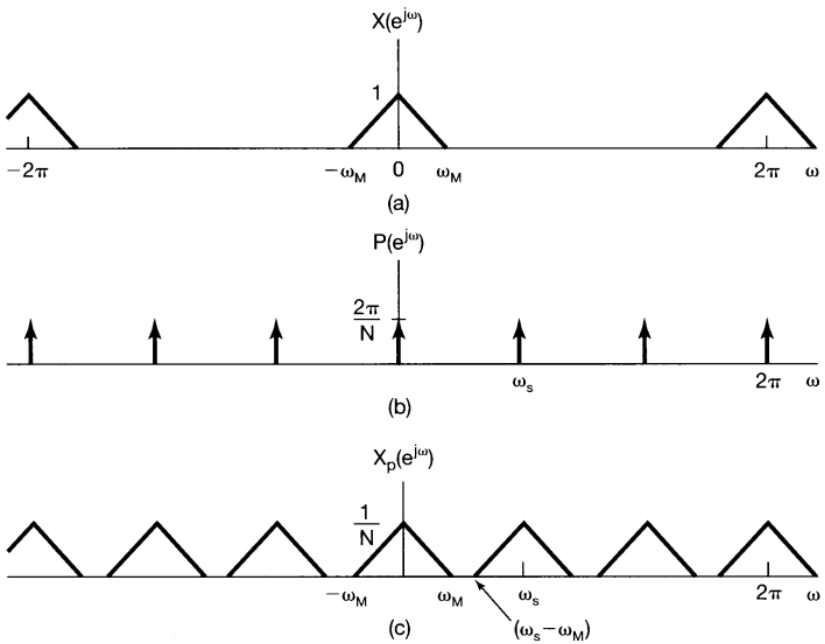


Image credit: Oppenheim 7.33

Sampling of discrete-time signals

Aliasing can happen in DT; some differences due to DT frequency range (π is the highest frequency).

Exercise: suppose $x[n]$ has $X(e^{j\omega})$ that is 0 for $3\pi/7 \leq |\omega| \leq \pi$. What is the largest sampling period N we can use without aliasing?

Decimation

Sampling and transmitting a DT signal like this is inefficient

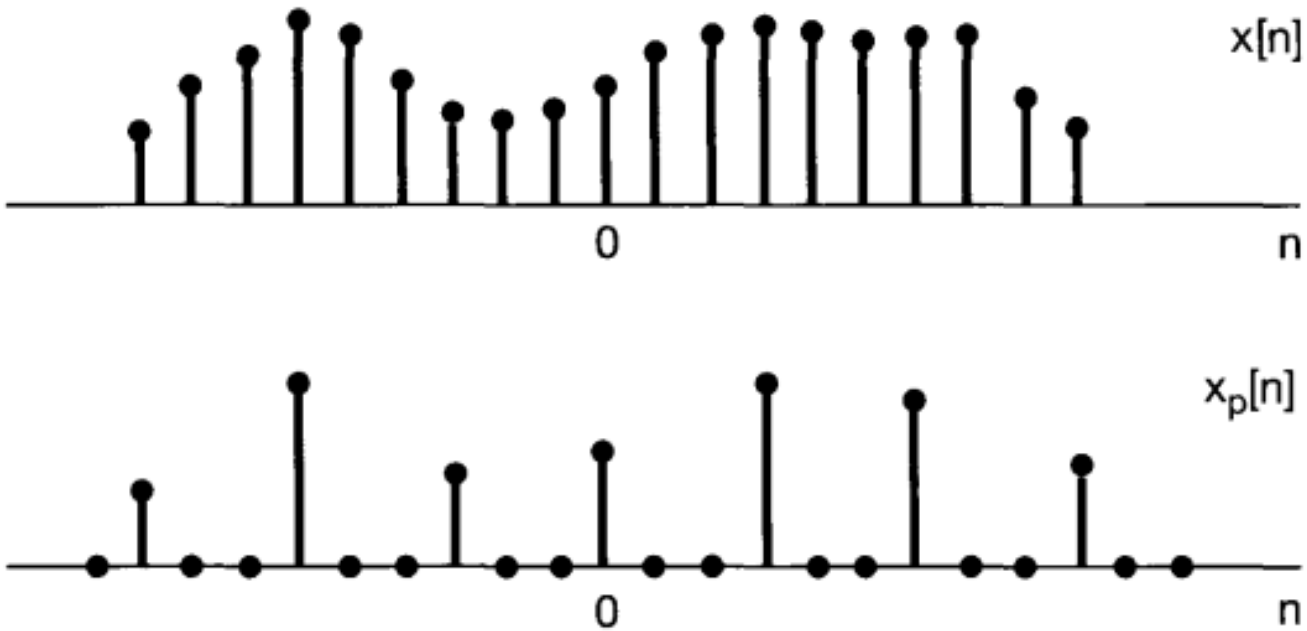


Image credit: Oppenheim 7.31

Decimation

We can *compress* the representation:

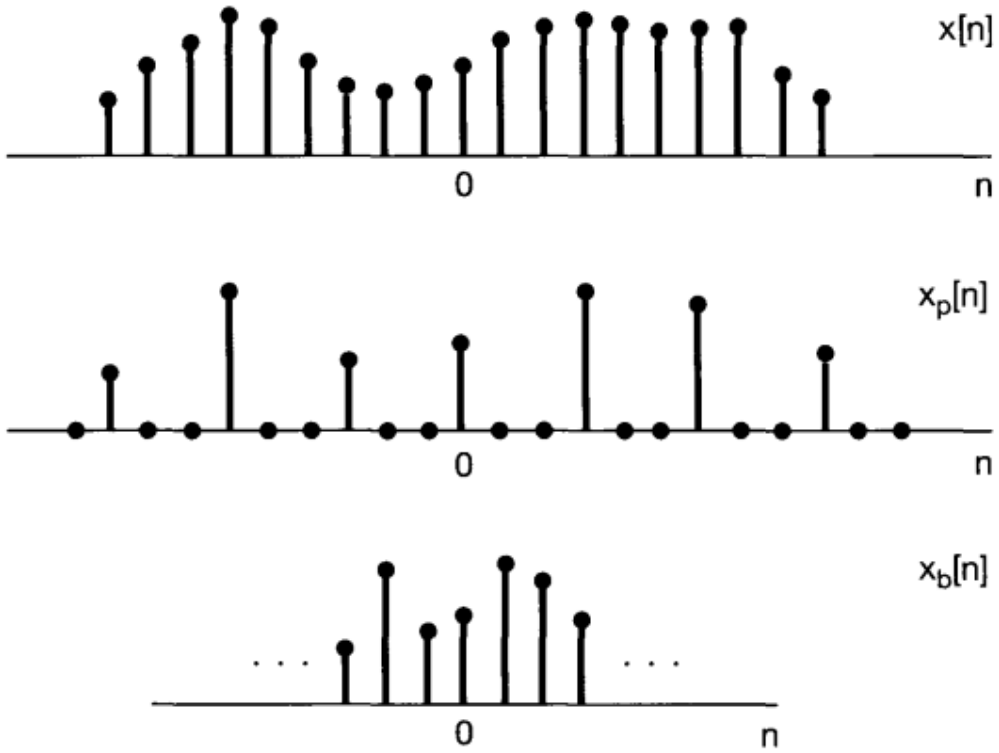


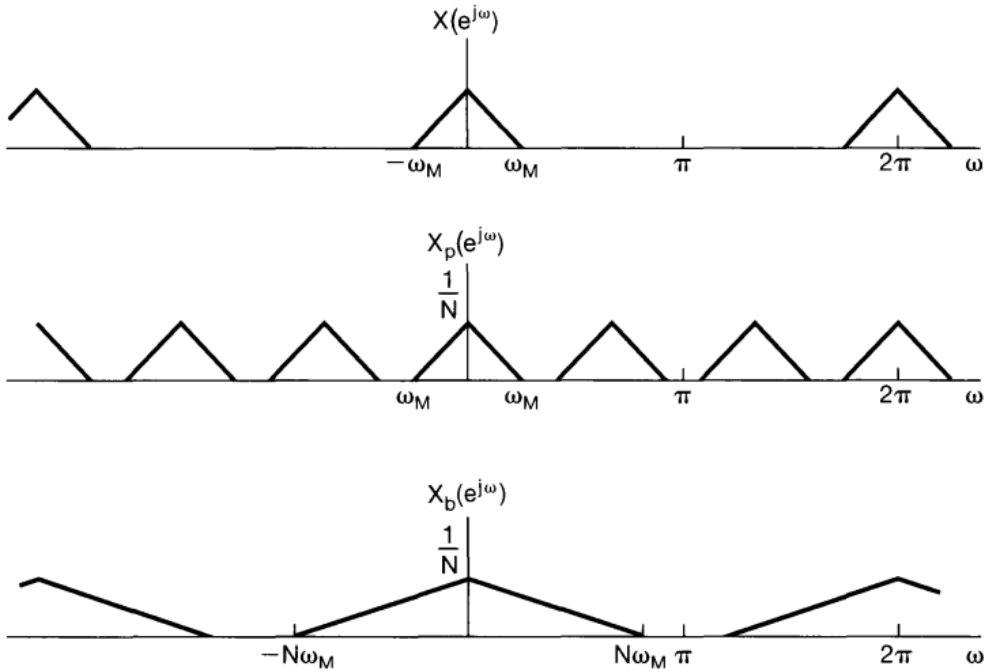
Image credit: Oppenheim 7.34

Decimation

Frequency domain effect:

Decimation

Decimation spreads out the spectrum



If original signal was CT, say that decimation has *downsampled* it.

Image credit: Oppenheim 7.35

Interpolation (upsampling)

Opposite of decimation: add $N-1$ points between.

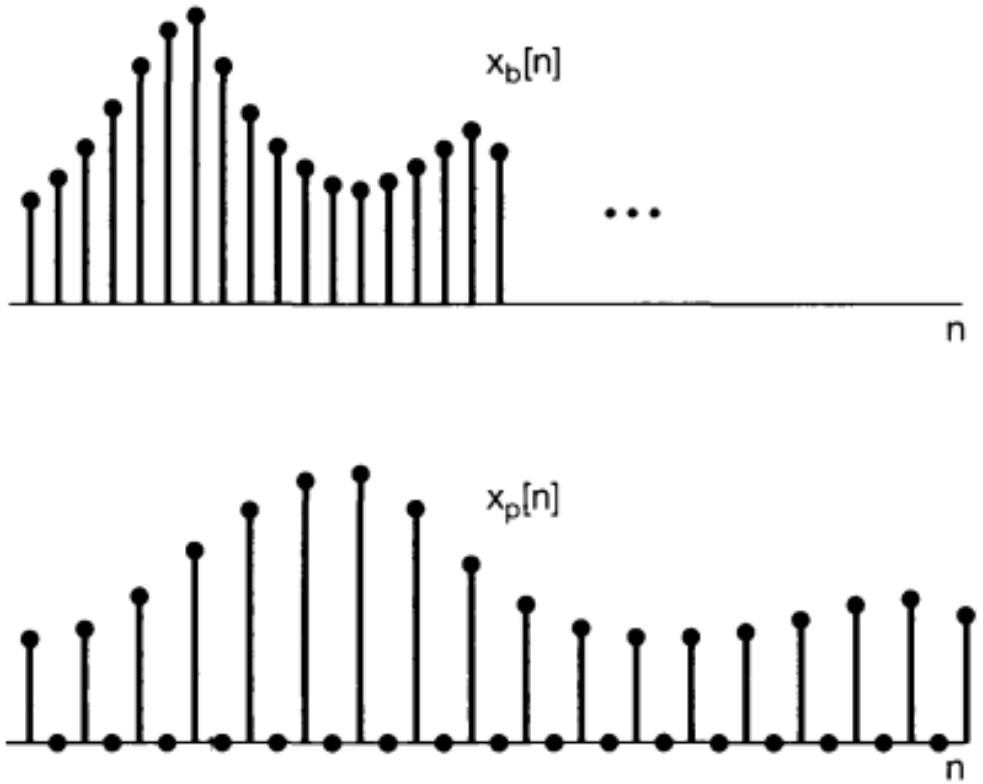


Image credit: Oppenheim 7.5

Interpolation (upsampling)

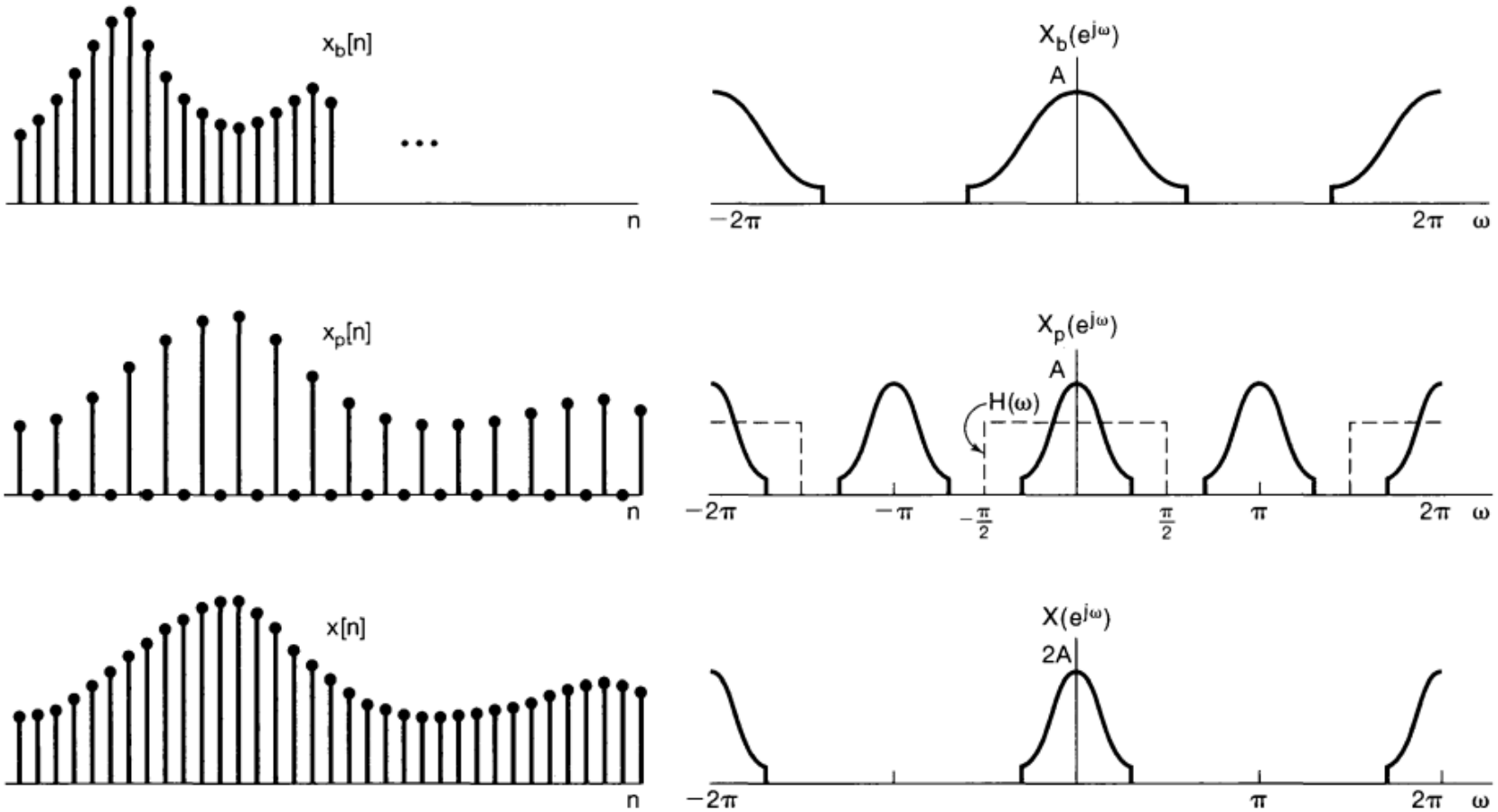


Image credit: Oppenheim 7.5

Example: down/upsampling

Oppenheim problem 7.19,

7.19. Consider the system shown in Figure P7.19, with input $x[n]$ and the corresponding output $y[n]$. The zero-insertion system inserts two points with zero amplitude between each of the sequence values in $x[n]$. The decimation is defined by

$$y[n] = w[5n],$$

where $w[n]$ is the input sequence for the decimation system. If the input is of the form

$$x[n] = \frac{\sin \omega_1 n}{\pi n},$$

determine the output $y[n]$ for the following values of ω_1:

(a) $\omega_1 \leq \frac{3\pi}{5}$

(b) $\omega_1 > \frac{3\pi}{5}$

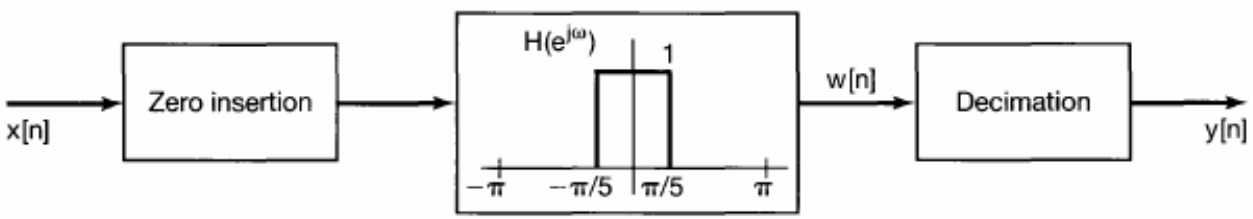


Figure P7.19

Example: down/upsampling

First case: $\omega_1 \leq \frac{3\pi}{5}$

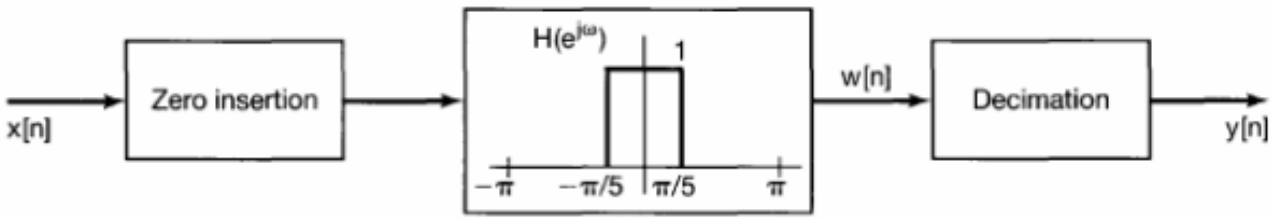


Figure P7.19

Example: down/upsampling

Second case: $\omega_1 > \frac{3\pi}{5}$

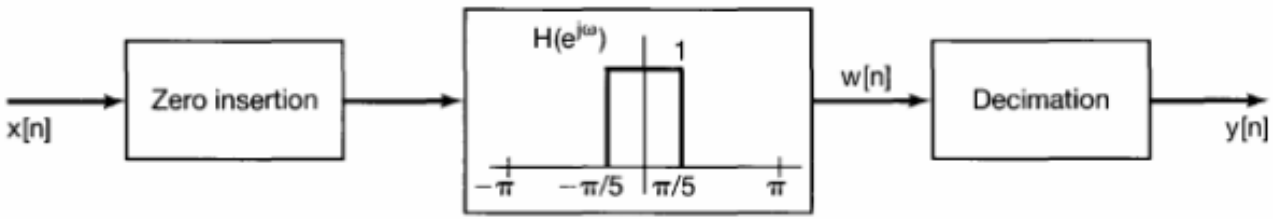


Figure P7.19

For next time

Content:

- new topic: modulation / communication systems

Action items:

1. Try questions in practice assignment 4
2. Prepare for quiz 8 on Tuesday (L17 and L18 material)
3. Tutorial Assignment 4 Monday

Recommended reading:

- From this class: Oppenheim 7.4-7.6
- Suggested problems: 7.17, 7.18, 7.20, 7.30, 7.32
- For next class: Oppenheim 8.0-8.4